

Problems Session 11

Strassen's Algorithm

Problem 1

How quickly can you multiply a $kn \times n$ matrix and a $n \times kn$ matrix using Strassen's algorithm as a subroutine? Answer the same question with the order of the matrices reversed.

Problem 2

Show how to multiply the complex numbers $a + ib$ and $c + id$ using only three multiplications of real numbers. The algorithm should take a, b, c and d as input and produce the real component $ac - bd$ and the imaginary component $ad + bc$.